

HOW TO CRASH A CARD GAME

HEY KEYNOTE LETS YOU MAKE SOME REALLY UGLY BACKGROUNDS

DISCLAIMER

I'm doing this because it's funny

You will learn absolutely nothing from this

WHAT IS YU-GI-OH

6. THE GAME MAT



THIS IS AN OLDER VERSION OF THE GAME

DO NOT ASK ME ABOUT LINK OR XYZ OR PENDULUM SUMMONING PLEASE I AM ASKING YOU SO NICELY

WHAT IS YU-GI-OH



MONSTER CARDS:

- HAVE NAMES
- HAVE NUMBERS
- CAN BE FACE-UP OR FACE-DOWN
- SOMETIMES HAVE CARD TEXT

WHAT IS YU-GI-OH



SPELL AND TRAP CARDS:

- **HAVE NAMES**
- **NO NUMBERS**
- **CAN BE FACE-UP OR FACE-DOWN**
- **ALWAYS HAVE CARD TEXT**

CARD TEXT

Determines what the card does when activated

(By default, “activated” = as soon as the card is face-up on the field)

A small function over board state written in natural language

They weren't 100% sure what they were doing.

“SMALL”

Needs to fit on the card, so there is some upper limit to complexity

... Albeit a high one

There is some syntax + reserved words (Problem Solving Card Text) but you can write anything here



“FUNCTION”

Most effects “happen” (or “resolve”) at a specific time

Except continuous-like effects

Continuous-like effects are always happening

(While the card is face-up on the field)



MOST IMPORTANTLY:

You have to have a favourite card

Ideally, it kind of sucks



MY FAVOURITE CARD:



POLE POSITION

MY FAVOURITE CARD:



POLE POSITION

**THE FACE-UP MONSTER(S) WITH THE
HIGHEST ATK ON THE FIELD IS
UNAFFECTED BY ANY SPELL CARDS.**

MY FAVOURITE CARD:



I'M ABOUT TO DO SOMETHING REALLY FUNNY



I have Celtic Guardian (1400ATK) face-up

My opponent has Gazelle the King of Mythical Beasts (1500ATK) face-up

Pole Position is in play, affecting Gazelle the King of Mythical Beasts



I'M ABOUT TO DO SOMETHING REALLY FUNNY

I equip **Legendary Sword** (+300ATK) to **Celtic Guardian**



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It is now the monster with the highest attack



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It is immune to the effect of **Legendary Sword**



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It is immune to the effect of **Legendary Sword**

It is no longer the monster with the highest attack



I'M ABOUT TO DO SOMETHING REALLY FUNNY

I equip **Legendary Sword** (+300ATK) to **Celtic Guardian**



It is now the monster with the highest attack

It is immune to the effect of **Legendary Sword**

It is no longer the monster with the highest attack

It is now affected by **Legendary Sword**



WHAT HAS HAPPENED HERE?



1. Yu-Gi-Oh game states are (at worst) countably infinite
2. The question “does this monster have the highest ATK value on this board” is a predicate expressible in card text
3. This predicate is neither true nor false
4. If you reason in the other direction, actually it’s both true and false

WHAT HAS HAPPENED HERE?



Assuming Celtic Guardian either does or does not have the highest attack:

- 1. If it does, then Pole Position is reducing its ATK by 300, therefore it does not
- 2. If it does not, then trivially it does not

[YES]	[NO]	YES V NO
NO	NO	
		$\vee_E$
NO		

WHAT HAS HAPPENED HERE?



Assuming Celtic Guardian either does or does not have the highest attack:

- 1. If it does, then trivially it does
- 2. If it does not, Legendary Sword is increasing its ATK by 300, therefore it does

[YES]	[NO]	YES V NO
YES	YES	
		$\vee_E$
YES		

WHAT DO WE DO ABOUT THIS?

OPTION 1: BAN ALL NON-TERMINATING CARDS

Bad solution!

- **Konami don't want to ban someone's favourite card**
- **Non-terminating effect sequences were accepted early on**
- **If you can construct a recursive combo, that's skillful play**

WHAT DO WE DO ABOUT THIS?

OPTION 2: THE GAME STALLS UNTIL THE TIME LIMIT

The player with the higher LP is declared the winner, if equal play a sudden death decider match

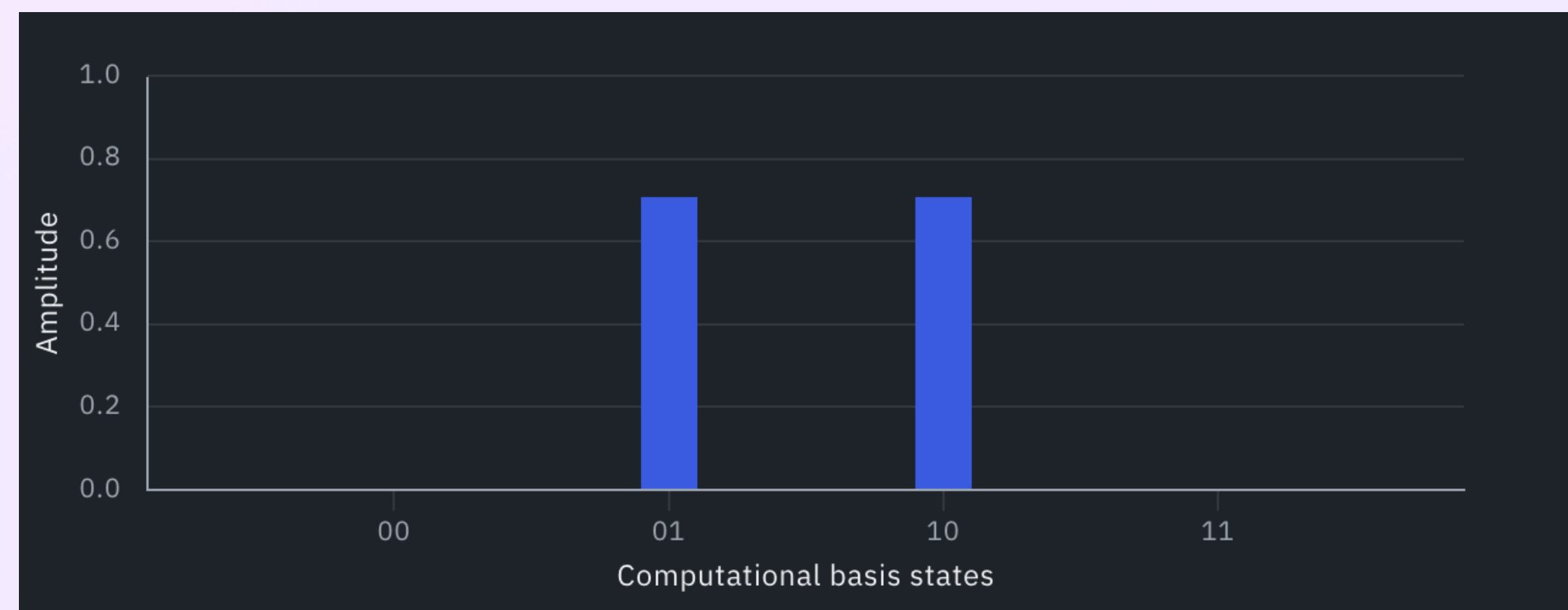
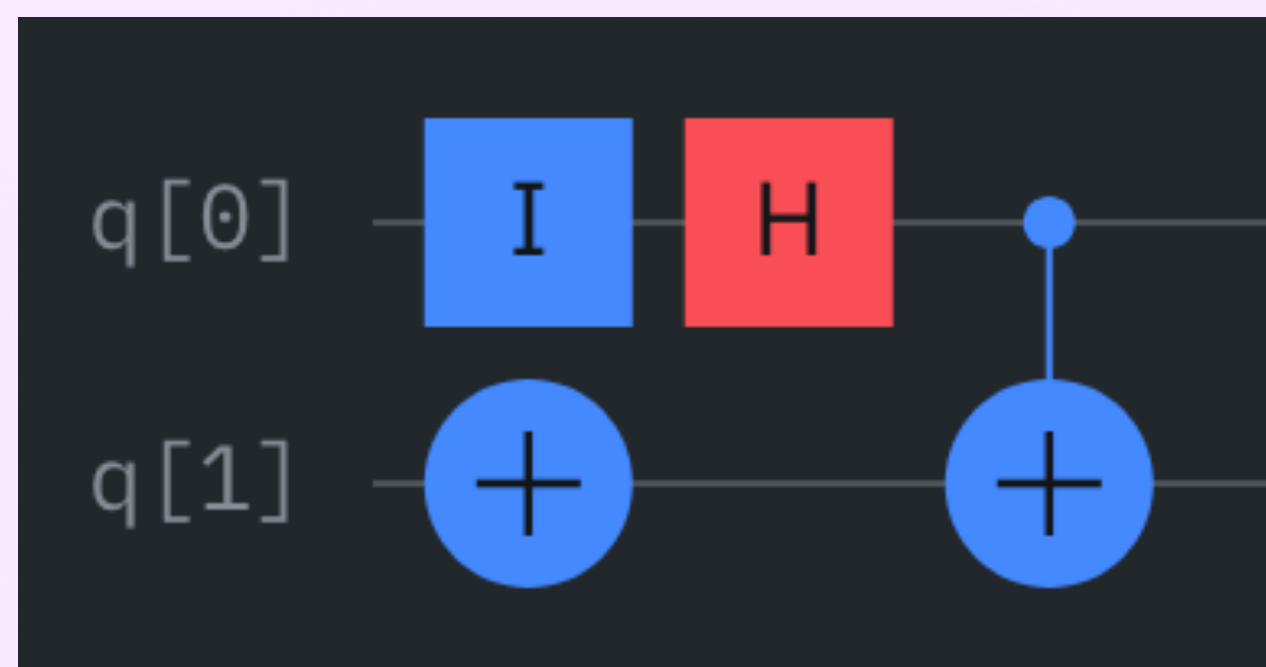
- **Cards which do this always get banned**
- **A single card is overpowered if it decides a whole game**

WHAT DO WE DO ABOUT THIS?

OPTION 3: QUANTUM GAME STATE

The game enters a superposition in which Celtic Guardian and Gazelle the King of Mythical Beasts both have the highest attack

- Every card effect terminates
- Need to explain quantum superposition to 12 year olds



WHAT DO WE DO ABOUT THIS?

OPTION 4: THE ACTUAL RULING

4. If no victory condition will be achieved, the Duelists should call a judge to determine the card that is the primary cause of the loop and send it to the Graveyard. Play will resume from that point and no penalties are assessed. Any effects that could activate as a result of the card going to the Graveyard can/will still activate accordingly.
 - If the judge determines that multiple cards primarily cause the loop, the card most recently activated should be sent to the Graveyard.

Official KDE-US Yu-Gi-Oh! TRADING CARD GAME Tournament Policy - V.2.2

Ask a tournament official to fix your game whenever this happens

WHAT DO WE DO ABOUT THIS?

Examples:	<u>Infinite Loop:</u> Duelist A has a Set Pole Position on the field. Duelist B has a face-up Dark Magician equipped with Magic Formula and a face-up Dark Magician Girl equipped with Axe of Despair. If Duelist A activates his Set Pole Position, its Spell protection is applied to Dark Magician, then to Dark Magician Girl then back to Dark Magician. The players determine this is an Infinite Loop that will not result in a victory condition, so they call a judge. The judge determines Pole Position is causing the loop. Pole Position is sent to the Graveyard, and play continues.
	<u>Infinite Loop:</u> Duelist A has a Set Pole Position on the field. Duelist B has a face-up Dark Magician equipped with Magic Formula, a face-up Dark Magician Girl equipped with Axe of Despair, and Bait Doll in his hand. During his Main Phase 1 he activated Bait Doll, targeting the Set Pole Position. Pole Position is forcibly activated. Its Spell protection is applied to Dark Magician, then to Dark Magician Girl then back to Dark Magician. The players determine this is an Infinite Loop that will not result in a victory condition, so they call a judge. The judge determines Pole Position is causing the loop. Pole Position is sent to the Graveyard, and play continues.

They will tell you to stop playing Pole Position.